

- Coriolis acceleration 

$$\vec{a}_p = \vec{a}_o + \frac{{}^b d^2}{dt^2} \vec{r} + \underbrace{2\vec{\omega}_{ib} \times \frac{{}^b d}{dt} \vec{r}}_{\text{Coriolis acceleration}} + \underbrace{\vec{\alpha}_{ib} \times \vec{r}}_{\text{Transversal acceleration}} + \underbrace{\vec{\omega}_{ib} \times (\vec{\omega}_{ib} \times \vec{r})}_{\text{Centripetal acceleration}} \quad (1)$$

- Transversal acceleration 
- Centripetal acceleration 